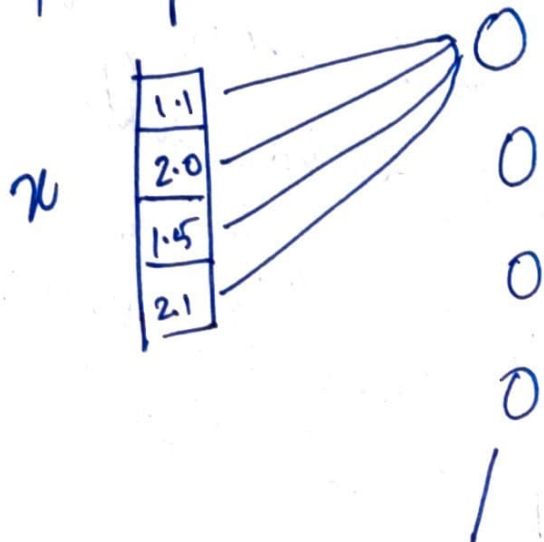


# Normalization

How to norm

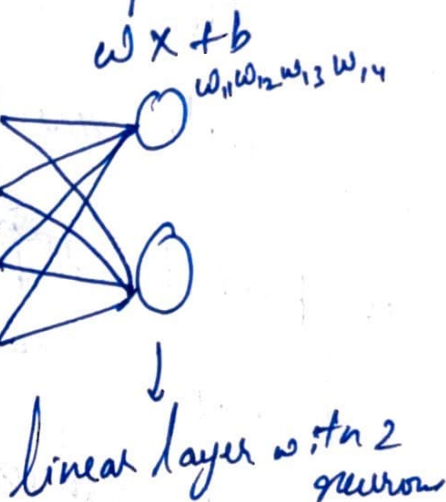
output features = 4  
input features = 4



$$w_{11} \quad w_{12} \quad w_{13} \quad w_{14}$$

$$x' = \begin{bmatrix} 1.6 \\ 2.7 \\ 1.1 \\ 3.1 \end{bmatrix}$$

input features = 4  
output features = 2



# neurons = # of output features = 2

$w \cdot x$   
dot product

Issue in this  $\rightarrow$  "Covariate Shift"  
when the input vector changes from 1 batch to other in magnitude, then the output of the layer also changes a lot depending on input vector. Hence the loss will also change a lot, leading to loss of gradient. Which will result in loss of oscillation, Model trains slowly.

## Solution

## Batch Normalization

In two, instead of having a single of list of images we will have batches of multiple inputs

- Calculate statistics of each item such as mean, Variance divide by standard deviation, this will be distributed according to Gaussian mean 0 and variance 1
- by this, model will not see much change when the inputs are different, say there are 2 different inputs a cat and zebra whose embeddings are very different, so here instead of 2 different values, due to math, the both are distributed with same calculation, hence the values are not very different but scaled. Model doesn't oscillate much, Training happens faster.

$$x^k = \frac{x^k - \text{Mean}}{\text{Standard dev}} \rightarrow [0, 1]$$

Stable Training.

- Better to use big batch size to avoid diversity issues